Cyclistic Bike-Share Analysis: How Casual Riders and Members Differ

Ask:

Cyclistic is a bike-share program located in Chicago that features a variety of bicycles and docking stations such as traditional bicycles, reclining bicycles, hand tricycles, and cargo bikes. This company assists people with disabilities and riders who cannot use a two-wheeled bike.

Cyclistic's marketing strategy relied on building general awareness which allowed members to either be a casual rider or an annual member. However, Cyclistic's finance analysts have concluded that annual members are much more profitable compared to the casual riders and for future growth. Hence, the clear goal of the company is to design marketing strategies that are aimed at converting casual riders to annual members and the question that I have been assigned to answer is "How do annual members and casual riders use Cyclistic bikes differently?".

In order to reach this goal, as a business we must examine the behavioral differences between the annual members and casual riders in order to identify trends that can possibly support membership conversion strategies.

Prepare:

The datasets that were used in this case study were taken from a data bucket named “divvy-tripdata”, and I took the 12 most recent months of data and started to analyze those (May 2025-April 2026). Each of these datasets were organized in the same format, each had a column for the ride ID, bike type, start and end time, start and end station name and id, start and end latitude and longitude, and member vs casual.

Instead of stating the names of the riders, each data entry had a ride ID which protected the rider’s privacy as their names weren’t leaked. The dataset appeared internally consistent and suitable for analysis, although the lack of rider-level identifiers limits the ability to study individual customer behavior over time. However, while the company is protecting their customers by not leaking their names or member IDs, this leads to one drawback as we are unsure if a rider has taken more than one ride on a single day.

Process:
In order to clean the data, I initially tried to manually clean each dataset by hand as I felt this was the tool that I was the most comfortable using and I would be able to track each change that was made to the dataset. However, after considering the lengthiness of these datasets and how arduous the task would really be, I wrote queries in BigQuery using SQL to properly clean the data and to assist with the analysis queries, I initially combined all of the datasets into one table named all_trips. Instead of writing queries for all of the datasets separately, I would write queries for each column to document each change that I made to the table.

Initially, I added three more columns to the all_trips table (ride_length, day_of_week, month). Ride length was calculated by using TIMESTAMP_DIFF between ended_at and started_at and then formatted by minute. The day of the week was calculated by extracting the day of week from started_at. The month was calculated by extracting the month from started_at. The initial combined dataset currently had 5,697,455 rows.

Then, in order to eliminate ride times that were invalid, I wrote a query that eliminated all rides that had a ride length less than 1 minute or greater than 1440 minutes. This needed to be done as it would be illogical for a ride to be less than one minute and the company cannot have rides for over 24 hours, which is 1440 minutes. After taking this step, the updated dataset would have 5,535,677 rows of data, meaning that this removed 161,778 rows of invalid data.

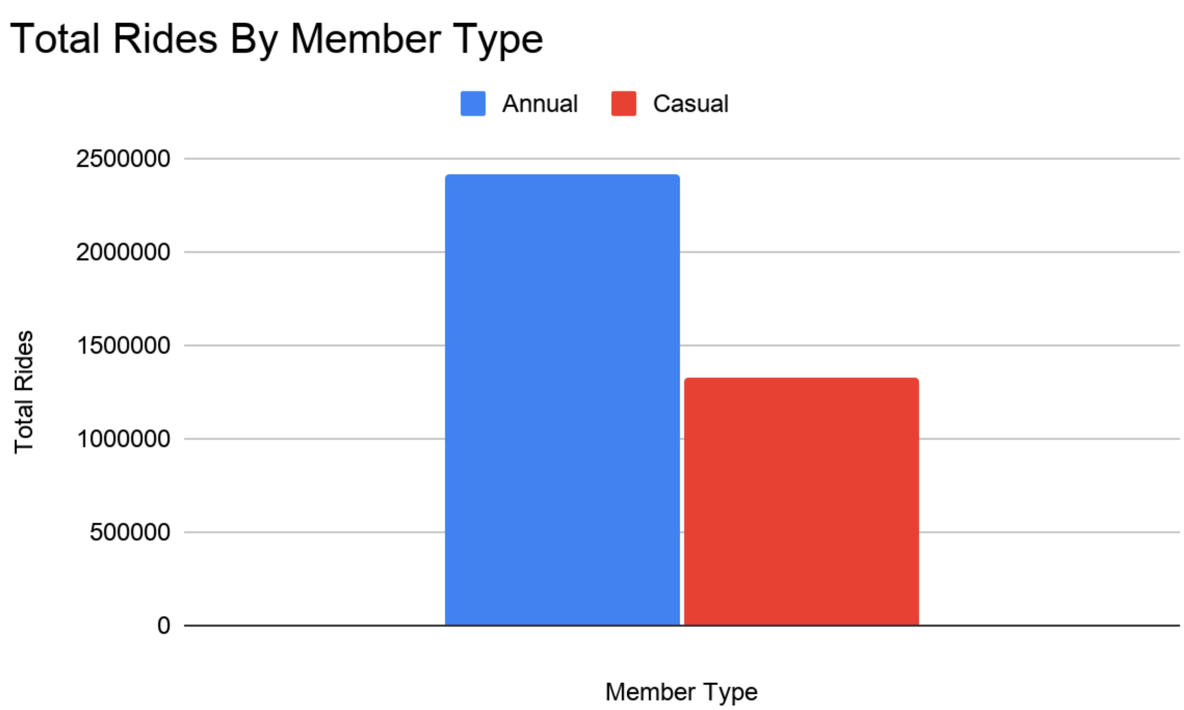
After getting rid of the entries with invalid ride lengths, I filtered out all entries where the name of the start station was empty and all entries where the name of the end station was empty as well as these are incomplete and invalid entries and ones that won't help us in the full picture. After filtering out invalid start station entries, the total rows went down to 4,414,611. After filtering out invalid end station entries, the total rows went down to 3,747,342. These two cleaning steps brought the total rows down by 1,788,335.

Finally, as the last step of the data cleaning, I filtered out entries that weren't of type member or casual as these could be considered as invalid and nowhere near what we are looking for with our analysis. This step was actually unnecessary to take as no rows were deleted and there weren't any invalid entries for the member type.

Analyze:

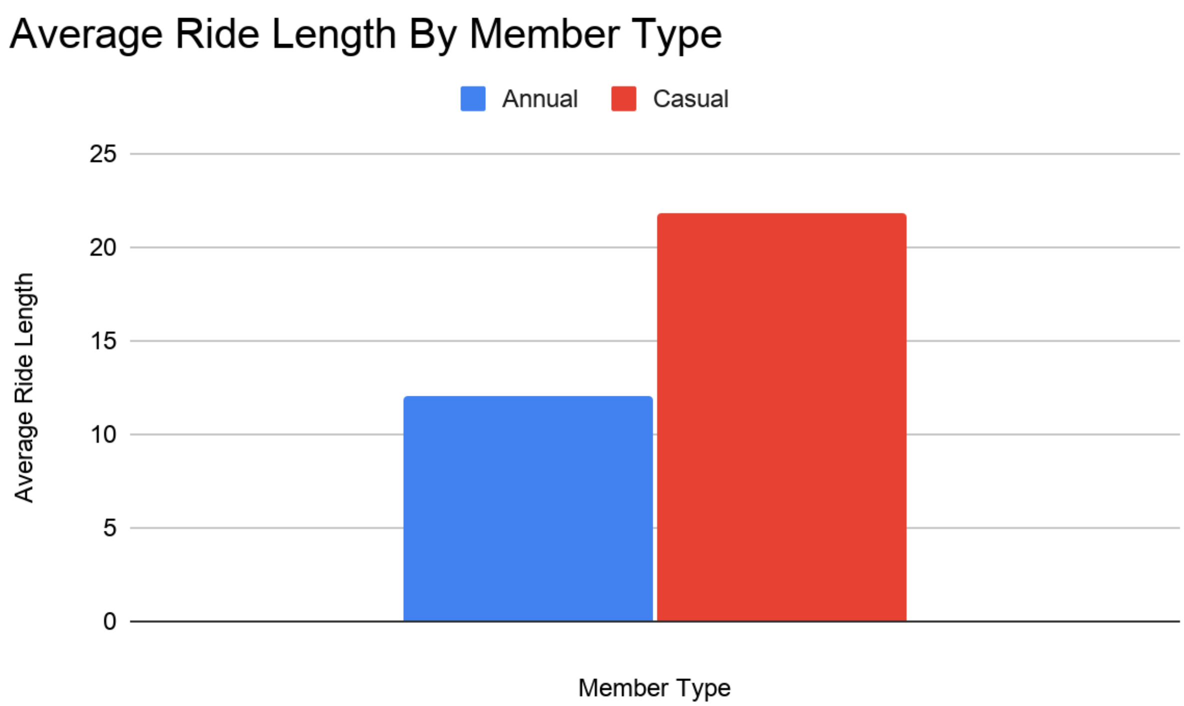
After writing the cleaning queries, I needed to start writing the analysis queries which would generate new sheets that I can create charts from. The charts that I created for my analysis were Total Rides By Member Type, Average Ride Length By Rider Type and Day of Week, Average Ride Length By Member Type, Rides By Rider Type and Day of Week, and Rides By Month by Member Type.

Over the past 12 months, the charts have made it clear that more members schedule rides compared to casual riders (2,418,896 members vs 1,328,356 casual riders).



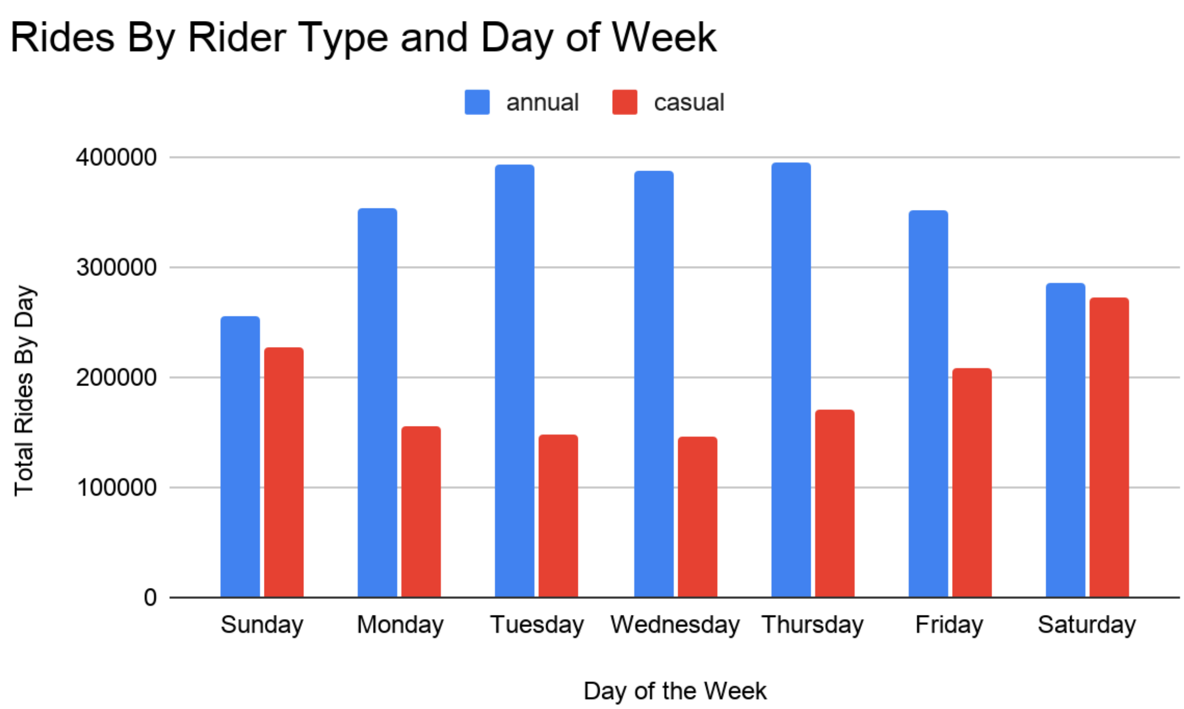
Annual members took 82% more rides than casual riders over the 12-month period.

However, the company's goal is to change these causal riders to members and make the casual riders count close to zero. Analyzing the data even more, it is clear that casual riders use their bikes for longer compared to members (21.7689 minutes vs 12.0479 minutes). Using this data, I can infer that casual riders spend more time on each ride but take fewer trips, which suggests that casual riders may use the service less frequently but for longer leisure-oriented trips.



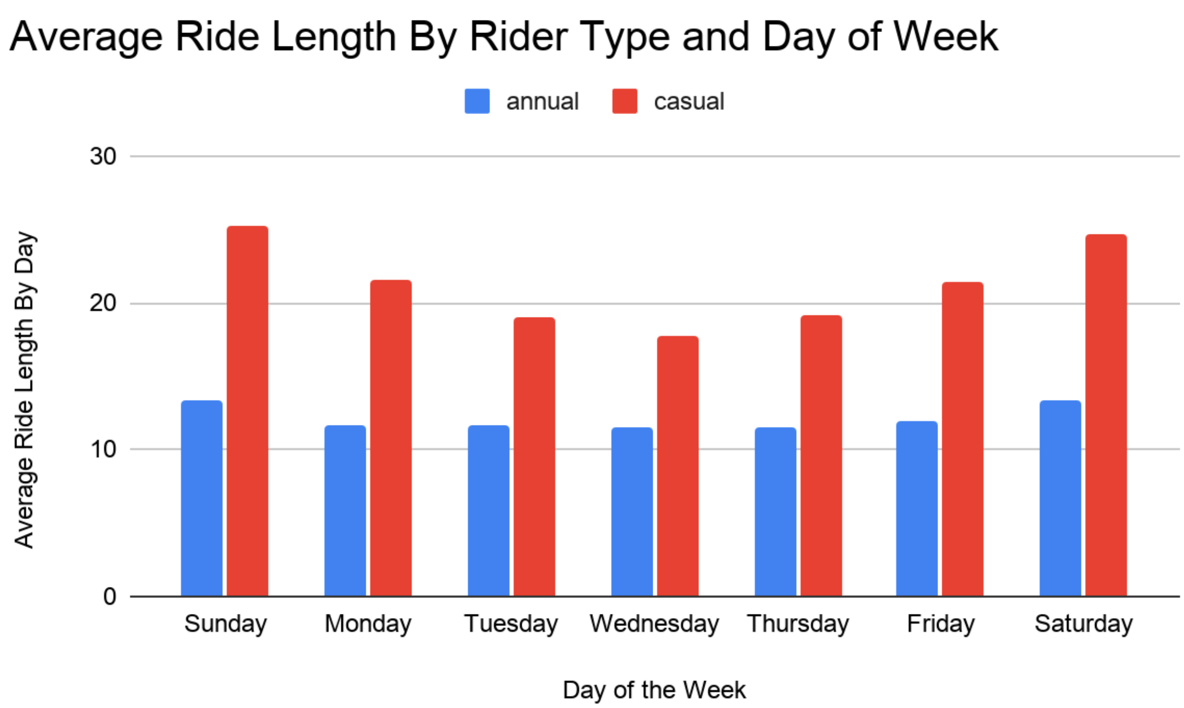
Casual riders average nearly twice the ride duration of annual members.

Examining the member vs casual rides deeper, I created a chart that compares the amount of rides for each member type for each day of the week. For each of the days, the total members who book rides are higher compared to the casual riders. On weekdays, the member count is significantly higher compared to casual riders. However, on weekends, the bars are almost level with lesser members and more casual riders. This signals that casual riders are more likely to book rides for more recreational purposes. As for the members, people who usually purchase the full-time membership are more likely going to use it as a daily commute to work, school, etc.



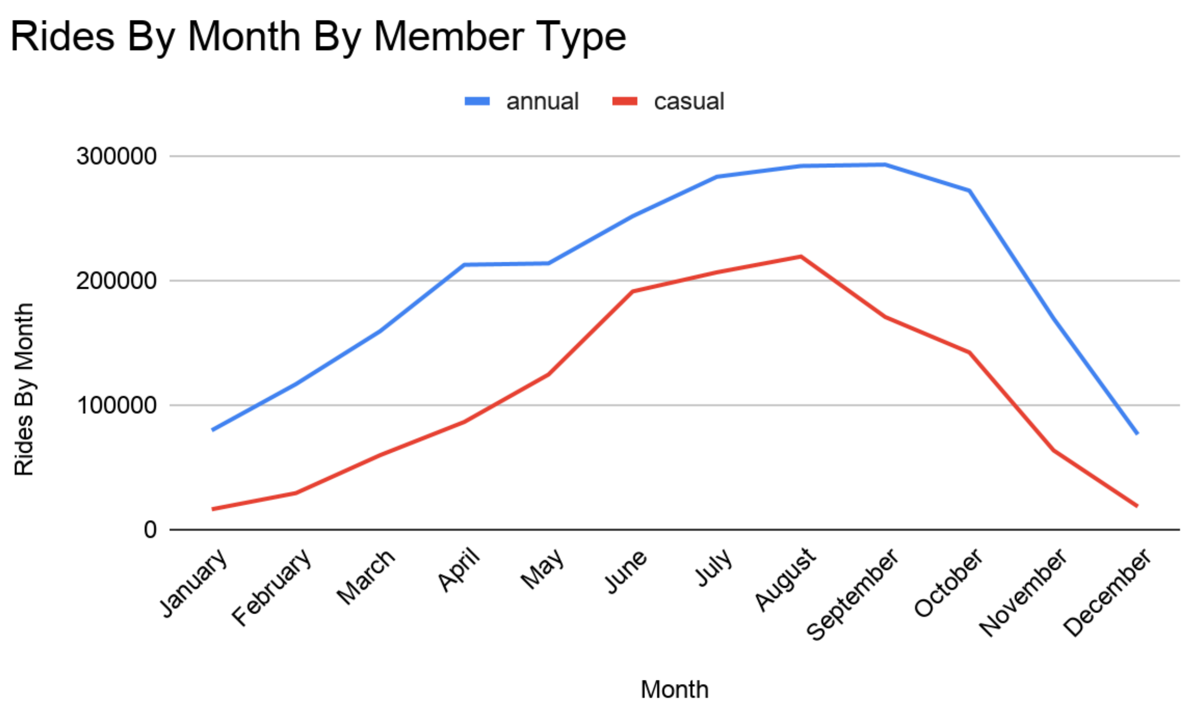
Members dominate weekday ridership; casual riders nearly match members on Saturdays.

Then, I compared the average ride length for members and casual riders once again and in this instance, I compared them by day of the week. The story was similar to the Average Ride Length By Member Type chart, the casual riders had the higher average ride length compared to members. Similar to the Rides By Rider Type and Day of Week chart, the counts for each type were higher on weekends compared to weekdays. However, for each day, the average ride length for a casual rider was more compared to a full-time member. Casual riders appear more likely to use bikes for leisure and recreational activities, especially during the weekends, whereas members appear to rely on bikes more consistently for commuting and routine transportation. Additionally, the higher average lengths on weekends signal that they use the bikes to travel to further places.



Casual riders consistently ride longer than members every day of the week.

Finally, I created a chart that divides the total rides for members and casual riders by month and unlike the other charts which were bar graphs, this was a line graph. For both lines, the total rides were lowest during the winter months (December, January, February) and highest during the summer months (June, July, August, September). The member line had a peak in September with 292,843 rides and the casual line had a peak in August with 219,136 rides. Taking in all of this information, ridership increases significantly during warmer months, suggesting seasonal weather influences bike usage.



Both groups peak in summer; casual ridership drops near zero in winter while members stay above 100,000 rides.

Act:

Based on all of this data and the analysis charts I was able to create, there are three recommendations that I am able to make in order to help the future growth of the company.

First, I believe that the company should create some pathways for the casual riders with single-ride and full-day passes into becoming full-time members. If a certain person purchases a specific number of passes from the company, discounts can be provided to them to become full-time members, which can help to sway their opinion on becoming a full-time member or not.

Another recommendation is on weekends, since the company gets more casual riders during that time period, have optional programs for the casual riders to explain to them the benefits of becoming an annual member.

Lastly, over the summer months, the company can add a new branch of flexibility such as a three-month pass which has the same benefits and opportunities as a full-time member can have at a discounted price. This way, a casual rider will understand what it's like to be a full-time member and can decide whether to switch or not after the three-month period is over.

Conclusion:

After analyzing all of the present data handed to us, annual members and casual riders exhibit different usage patterns. While members ride more frequently and consistently throughout the week which suggests commuter and routine transportation usage, casual riders take fewer but longer trips and show increased activity during the weekends and summer months which indicate recreational usage. These differences create opportunities for targeted marketing campaigns, seasonal membership offerings, and conversion incentives aimed to increase annual memberships.